

KANOYA AB, Japan

WMO: 478500

Lat: 31.368N

Lon: 130.838E

Elev: 224

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.3	(c) 33.4	(d) 19.6	(e) 14.9	(f) 39.7	(g) 22.7	(h) 17.2	(i) 40.4	(j) 28.6	(k) 50.4	(l) 25.9	(m) 51.2	(n) 3.1	(o) 20	(p) 0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.6	(c) 90.0	(d) 79.0	(e) 89.4	(f) 78.9	(g) 87.8	(h) 78.8	(i) 80.9	(j) 86.6	(k) 80.0	(l) 85.6	(m) 79.4	(n) 84.7	(o) 11.4	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	152.4	84.2	78.8	150.8	83.9	77.4	143.9	82.8	44.7	87.9	43.9	86.7	43.2	85.1	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.5	(b) 21.9	(c) 19.2	(e) 25.2	(f) 92.8	(g) 2.4	(h) 1.9	(i) 23.4	(j) 94.1	(k) 22.0	(l) 95.2	(m) 20.6	(n) 96.3	(o) 18.8	(p) 97.7		
(4) 25.5	(5) 21.9	(6) 19.2	(7) 23.4	(8) 82.4	(9) 2.2	(10) 0.8	(11) 21.8	(12) 83.0	(13) 20.6	(14) 83.4	(15) 19.4	(16) 83.8	(17) 17.8	(18) 84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	46.1	48.4	53.2	61.1	68.4	73.8	81.3	82.2	77.7	69.0	59.1	49.9	(6)
(7)		DBStd	13.42	5.45	6.31	5.66	5.16	3.70	3.93	2.88	2.27	3.87	5.31	6.31	6.25	(7)
(8)		HDD50	363	147	99	31	1	0	0	0	0	0	0	4	81	(8)
(9)		HDD65	2250	585	465	365	135	13	0	0	0	0	23	194	470	(9)
(10)		CDD50	5572	27	54	131	334	569	715	969	999	830	590	277	77	(10)
(11)		CDD65	1985	0	0	0	18	117	265	504	534	380	148	18	1	(11)
(12)		CDH74	16836	0	0	0	38	398	1430	5264	5840	3119	708	39	0	(12)
(13)		CDH80	4994	0	0	0	1	20	202	1762	2107	820	81	1	0	(13)
(14)	Wind	WSAvg	8.4	8.4	8.9	9.4	9.2	8.8	8.9	8.8	8.4	7.8	7.5	6.8	7.7	(14)
(15)	Precipitation	PrecAvg	95.6	3.3	4.4	6.3	7.5	8.1	21.4	13.1	9.3	9.6	4.6	4.3	3.5	(15)
(16)		PrecMax	152.3	7.7	8.7	13.4	16.6	14.0	43.1	39.9	24.7	20.9	14.7	8.7	7.4	(16)
(17)		PrecMin	61.6	0.7	1.7	1.9	2.3	2.2	8.5	1.9	2.2	1.5	0.8	1.9	0.4	(17)
(18)		PrecStd	19.2	1.5	2.1	2.4	3.5	3.2	7.5	7.9	5.2	5.7	3.3	1.9	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	69.6	71.6	77.4	82.4	87.4	92.0	93.0	89.8	84.6	77.4	69.7	(19)
(20)			MCWB	60.3	63.8	62.1	66.5	69.3	78.7	79.7	79.7	78.1	74.4	69.2	63.2	(20)
(21)		2%	DB	62.5	66.1	68.3	74.9	79.2	84.4	90.0	91.2	87.9	82.3	74.9	66.2	(21)
(22)			MCWB	57.3	61.3	60.3	64.3	67.4	76.2	79.1	79.3	78.1	72.9	67.0	60.0	(22)
(23)		5%	DB	59.2	62.9	66.3	72.2	77.4	82.4	89.3	89.6	86.2	80.3	71.9	64.0	(23)
(24)			MCWB	53.0	57.3	59.3	63.3	67.1	75.2	78.9	79.0	77.0	71.2	64.6	57.7	(24)
(25)		10%	DB	57.0	60.5	64.1	70.2	76.6	80.5	87.7	88.0	84.4	77.6	69.9	61.0	(25)
(26)			MCWB	50.7	54.7	57.5	62.6	66.9	74.7	78.7	79.0	75.8	69.5	63.4	54.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	65.3	65.7	71.0	75.6	79.8	81.9	81.9	80.8	77.9	72.7	65.0	(27)
(28)			MCD B	64.9	67.9	68.7	74.1	77.8	84.0	88.0	87.4	86.1	82.1	76.0	68.0	(28)
(29)		2%	WB	58.6	61.9	63.4	68.9	73.0	78.7	80.8	81.0	79.4	75.8	69.2	61.5	(29)
(30)			MCD B	61.5	65.1	66.5	71.9	75.9	82.2	86.6	86.8	84.5	79.9	72.6	64.9	(30)
(31)		5%	WB	54.7	59.0	61.1	66.7	71.1	77.9	80.0	80.2	78.6	74.0	66.8	58.8	(31)
(32)			MCD B	58.4	62.4	64.8	70.1	74.4	81.0	85.5	85.9	83.6	78.0	70.5	62.5	(32)
(33)		10%	WB	51.2	55.3	58.8	64.3	69.5	76.5	79.3	79.6	77.8	71.7	64.6	55.9	(33)
(34)			MCD B	55.6	58.8	63.0	68.8	73.2	79.3	84.4	85.3	82.6	76.5	68.8	59.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	16.5	16.3	15.7	14.6	10.4	11.0	11.6	12.6	15.0	16.5	16.8	(35)
(36)			MCDBR	19.3	18.1	18.6	18.5	17.5	12.6	13.7	13.6	13.5	15.7	17.2	18.2	(36)
(37)			MCWBR	14.9	14.0	12.7	10.7	9.0	6.4	5.9	5.7	6.3	8.4	10.9	13.1	(37)
(38)		5% WB	MCD BR	16.8	15.4	14.8	11.9	11.1	9.0	11.2	11.4	11.1	11.2	13.0	14.9	(38)
(39)			MCWBR	15.2	14.1	12.4	9.8	8.3	5.8	5.6	5.5	6.0	7.4	10.4	12.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.421	0.459	0.514	0.533	0.539	0.538	0.519	0.510	0.497	0.469	0.428	0.412	(40)	
(41)		taud	2.132	2.054	1.912	1.922	1.976	2.066	2.152	2.176	2.169	2.166	2.204	2.165	(41)	
(42)		Ebn at Noon	247	249	245	246	245	244	248	249	247	244	243	242	(42)	
(43)		Edh at Noon	41	48	59	61	58	53	49	47	45	43	37	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	779	971	1287	1501	1553	1273	1723	1725	1417	1165	873	736	(44)	
(45)		RadStd	68	90	90	151	173	199	188	115	150	132	87	70	70	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page